

Ship Auto Way Maps Points Tracking

Dokumentasi sistem **Ship Auto Way Maps Points Tracking** — kontrol kapal model ESP32-S3 dengan waypoint, telemetry real-time, NMPC di mini PC, dan dashboard PySide6.

Versi dokumen: dashboard 1.6 + Remote/User-05 + Cpp 2.0

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Dokumen dashboard: `README Local Monitor Dashboard-beta1.6.md`

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1. Ringkasan sistem

Sistem ini memungkinkan:

- **Monitoring live** posisi, yaw kompas, rudder, RPM, baterai, status Mini PC, serta surge `u` / sway `v` (dihitung di dashboard)
- **Perencanaan waypoint** di peta (Home + hingga 10 waypoint navigasi)
- **Pengiriman waypoint** ke kapal via ESP-NOW (`0xA1`)
- **Kontrol auto alg 2** — rudder dari NMPC mini PC (`timestamp,result`)
- **Kontrol auto alg 1** — waypoint + PD (opsional, compile-time)
- **Shutdown mini PC** dari dashboard (ESP-NOW `0xA2`, tanpa Wi-Fi laptop↔mini PC)
- **Analisis log CSV** — replay data dengan peta dan plot

Alur end-to-end:

```
Dashboard beta 1.6 (PySide6)
| USB serial 115200
▼
User-Side-05 (ESP-Now_ESP32-S3_User-Side-05)
| ESP-NOW peer-to-peer
▼
Remote-Side-05 (ESP-Now_ESP32-S3_Remote-Side-05) – di kapal
| USB Serial 115200
▼
Mini PC – Cpp_ReadWriteSerial-2.0-ENU-NMPC-beta
```

2. Komponen & path proyek

Komponen	Path (dari root repo)
Dashboard	Pythonfile/Way_Points_Tracking/Local Monitor Dashboard-beta1.6.py
User-Side	PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_User-Side-05/
Remote-Side	PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_Remote-Side-05/
Mini PC	Cpp_Files/Cpp_ReadWriteSerial-2.0-ENU-NMPC-beta/
Dokumen ini	Pythonfile/Way_Points_Tracking/Ship Auto Way Maps Points Tracking.md

Cpp_ReadWriteSerial-2.1-ENU-NMPC-beta adalah cabang yang menghitung u, v dari GPS. Stack di dokumen ini memakai **2.0** ($v = 0$, $u_0 = 0.6114$ m/s).

3. Arsitektur & alur data

3.1 Telemetry (kapal → laptop)

1. Remote baca sensor ~10 Hz → struct `DatatoSend` (64 byte, 24 field)
2. ESP-NOW ke User-Side
3. User-Side cetak CSV 24 kolom ke USB
4. Dashboard parse → Live / log / plot, lalu hitung u, v lokal

3.2 Waypoint (laptop → kapal → mini PC)

1. Dashboard: `$WPSET,...`
2. User-Side → ESP-NOW `0xA1` → Remote simpan RAM + cetak `[WP] ...`
3. User-Side balas `$WACK,OK` / `$WACK,ERR,...`
4. Mini PC 2.0 memakai `[WP]` Home sebagai origin ENU dan `[WP] #n` sebagai target

3.3 Rudder NMPC (auto alg 2)

1. Remote (CH6 auto) kirim CSV 8 kolom ke mini PC
2. 2.0 memakai `timestamp`, `lat`, `lon`, `yaw`, `yaw_rate`, dan daftar `[WP]`

3. 2.0 mengabaikan `calc_deg_servo_*` dan `gyro_z`
4. Mini PC kirim `$HB` (~1 Hz) + `timestamp,result` (offset rudder, derajat)
5. Remote set `mini_pc_link` dari heartbeat; pakai `result` sebagai offset rudder

ψ , bearing, `u`, dan `v` tidak dikirim balik ke Remote.

3.4 Shutdown mini PC

1. Dashboard tombol **Shutdown** (hanya jika Mini PC CONNECTED)
2. `$SHUTDOWN` → User → ESP-NOW `0xA2` → Remote → Serial `$SHUTDOWN`
3. `Cpp_ReadWriteSerial-2.0` jalankan `shutdown /s /t 5`
4. User balas `$SACK,OK` (forward ESP-NOW sukses — bukan konfirmasi OS mati)

4. Remote-Side-05

Proyek: `ESP-Now_ESP32-S3_Remote-Side-05`

- Sensor, actuator, RC PPM, waypoint RAM, telemetry ESP-NOW 24 kolom
- Yaw kapal: raw JY901 (−180...180) → wrap 0...360 → +90° pasang → `360 - yaw` = kompas CW (0 Utara, 90 Timur, 270 Barat)
- USB Serial ke mini PC: CSV 8 kolom, `[WP]`, `$SHUTDOWN`; terima `$HB + timestamp,result`
- Default `#define AUTO_TRACK_ALG 2` (mini PC / NMPC)
- Alg 2: `heading_setpoint` = salinan yaw, `heading_error` = 0
- Alg 1: `heading_setpoint` = bearing ke waypoint aktif

Detail: `PlatformIO/.../Remote-Side-05/src/README.md`

5. User-Side-05

Proyek: `ESP-Now_ESP32-S3_User-Side-05`

- Gateway USB ↔ ESP-NOW. Tidak terhubung ke mini PC
- Forward `$WPSET` → `0xA1`, `$SHUTDOWN` → `0xA2`
- CSV 24 kolom ke dashboard. yaw dan `heading_setpoint` diteruskan mentah (×100)

Detail: `PlatformIO/.../User-Side-05/src/README.md`

6. Dashboard beta 1.6

File: `Local Monitor Dashboard-beta1.6.py`

- Live: mode, Mini PC CONNECTED/DISCONNECTED, warning auto tanpa mini PC
- Tombol **Shutdown** sebelah status Mini PC (enable jika Connect + `mini_pc_link=1`)
- Map Points: Home + waypoints, **Send Way Points** (`$WPSET` / `$WACK`)
- Live: **u surge**, **v sway** (hitung lokal; bukan dari firmware)

- Logging & Analyze: CSV 24 kolom ditambah `u (m/s)`, `v (m/s)`
- Plot Heading Setpoint pada alg 2 menempel pada yaw

Detail: `README Local Monitor Dashboard-beta1.6.md`

7. Mini PC — Cpp_ReadWriteSerial-2.0

Path: `Cpp_Files/Cpp_ReadWriteSerial-2.0-ENU-NMPC-beta/`

Arah	Isi
Terima	CSV 8 kolom saat CH6 auto; <code>[WP] Home / [WP] #n; \$SHUTDOWN</code>
Pakai	<code>timestamp, lat, lon, yaw, yaw_rate</code> , daftar WP
Abaikan	<code>calc_deg_servo_1/2, gyro_z</code>
Kirim	<code>\$HB</code> tiap 1 s; <code>timestamp,result</code> (rudder °, ±45)

State NMPC: `v = 0`, `u0 = 0.6114 m/s`, $\psi = \pi/2 - \text{yaw}$, `r` dari `yaw_rate` (tanda minus). `L = 1.0107 m`.

Detail: `Cpp_Files/Cpp_ReadWriteSerial-2.0-ENU-NMPC-beta/README.md`

8. Protokol serial (Dashboard ↔ User-Side)

Arah	Format
User → PC	CSV 24 kolom telemetry
PC → User	<code>\$WPSET,<home_lat>,<home_lon>,<count>,...</code>
User → PC	<code>\$WACK,OK / \$WACK,ERR,<reason></code>
PC → User	<code>\$SHUTDOWN</code>
User → PC	<code>\$SACK,OK / \$SACK,ERR,<reason></code>

Baud: **115200**.

Serial Remote ↔ mini PC terpisah (bukan lewat User-Side):

```
timestamp,lat,lon,calc_deg_servo_1,calc_deg_servo_2,yaw,gyro_z,yaw_rate
[WP] Home: lat, lon
[WP] #1: lat, lon
$HB
timestamp,result
$SHUTDOWN
```

9. Protokol ESP-NOW

msg_type	Payload	Arah	Fungsi
(telemetry)	DatatoSend 64 B	Remote → User	Telemetry 24 field
0xA1	waypoints_payload ~180 B	User → Remote	Waypoint + home
0xA2	pc_command_payload 4 B	User → Remote	Perintah mini PC (cmd=1 shutdown)

Catatan: 0xA2 di versi 05 = shutdown mini PC (bukan tuning NVS dokumen lama).

10. Telemetry 24 kolom

Urutan sama di Remote DatatoSend, User CSV, dan dashboard:

1 timestamp, 2 lat, 3 lon, 4 speedMps×100, 5–6 servo×100, 7 yaw×100,
8 hdg_sp×100, 9 hdg_err×100, 10 rudder_cmd×100, 11 track_wp_index,
12 distance_to_wp×10, 13–18 IMU×100, 19–20 RPM, 21–22 battery×100,
23 mode_auto, **24 mini_pc_link**

mode_auto: 0 Manual, 1 Auto PD, 2 Auto Mini PC.

Log dashboard menambah u (m/s) dan v (m/s) setelah speedMps. Kedua kolom itu tidak ada di firmware.

11. Yaw, setpoint, u dan v

Yaw hanya diubah di Remote:

```
raw (-180...180) → wrap 0...360 → +90° → 360 - yaw
```

Hasil di CSV: 0 = Utara, 90 = Timur, 270 = Barat. User-Side dan dashboard hanya ÷100.

Heading setpoint bukan mode zigzag. Pada alg 2 isinya salinan yaw. Bearing ke waypoint hanya diisi jika firmware di-compile dengan `AUTO_TRACK_ALG 1`.

u , v di dashboard 1.6 (kompas CW):

$$\begin{aligned}u &= \dot{x} \sin\psi + \dot{y} \cos\psi \\v &= \dot{x} \cos\psi - \dot{y} \sin\psi\end{aligned}$$

2.0 tidak memakai rumus ini.

12. Algoritma auto track

Dipilih compile-time di Remote (`AUTO_TRACK_ALG`):

Nilai	Perilaku
1	Waypoint haversine + PD rudder. <code>heading_setpoint</code> = bearing
2 (default)	Rudder dari mini PC <code>timestamp,result</code> . <code>Setpoint</code> = yaw

CH6 ≥ 1750 = Auto; jika alg 2 dan `mini_pc_link=0` $\rightarrow$ rudder netral + warning.

CH6 manual $\leftrightarrow$ auto tidak mereset indeks waypoint. Reset ke WP1 hanya lewat **Send Way Points** baru atau restart program 2.0.

13. Build, upload & menjalankan

```
# Firmware
cd PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_Remote-Side-05
pio run --target upload
cd ../ESP-Now_ESP32-S3_User-Side-05
pio run --target upload

# Mini PC 2.0
cd Cpp_Files/Cpp_ReadWriteSerial-2.0-ENU-NMPC-beta
g++ -std=c++17 -Iinclude -Inmpc src/main.cpp src/serial_port.cpp nmpc/nmpc_kapal_waypoint.c nmpc/read_write_serial.exe --port COMx --baud 115200 --rudder-mode nmpc --print all

# Dashboard
cd Pythonfile/Way_Points_Tracking
python "Local Monitor Dashboard-beta1.6.py"
```

Sesuaikan MAC ESP-NOW di kedua `main.cpp` dan port COM.

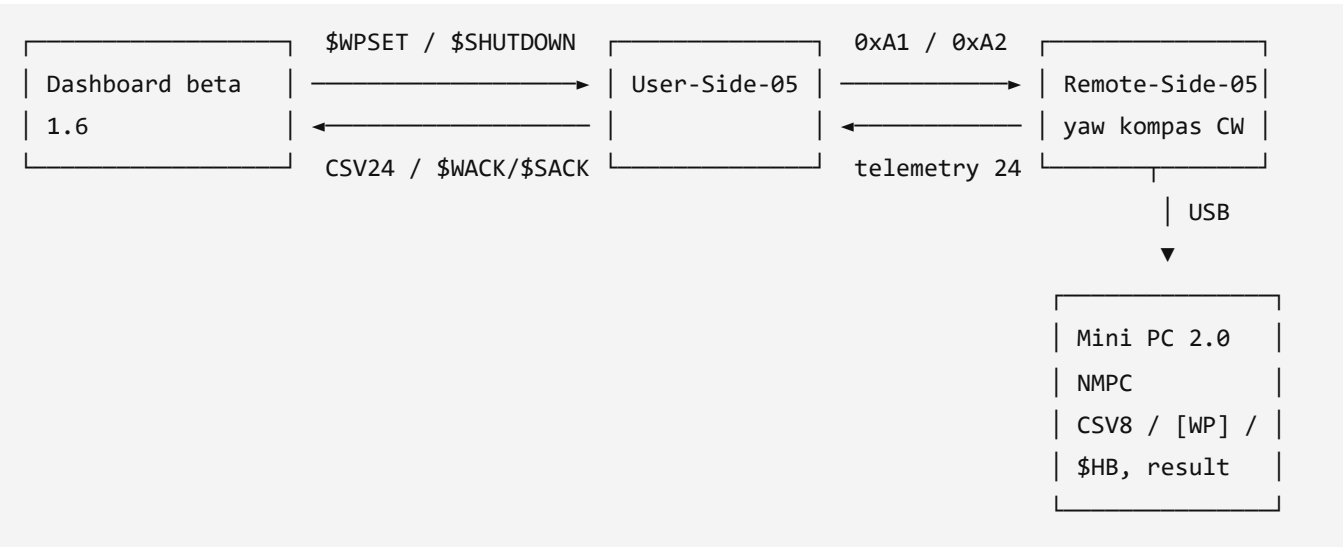
14. Prosedur uji lapangan

1. Flash **Remote-05** (rantai yaw CW) + **User-05** berpasangan
2. Jalankan **read_write_serial.exe** 2.0 di mini PC (auto-start opsional)
3. Connect dashboard beta **1.6** ke User-Side
4. Verifikasi Live: telemetry + Mini PC **CONNECTED**
5. Map Points $\rightarrow$ Set Home + ≥ 1 WP $\rightarrow$ **Send Way Points** $\rightarrow$ `$WACK,OK`; cek `[WP]` di stderr/stdout 2.0
6. RC CH6 Auto $\rightarrow$ pantau rudder dari `timestamp,result`
7. (Opsional) **Shutdown** $\rightarrow$ konfirmasi $\rightarrow$ `$SACK,OK` $\rightarrow$ mini PC mati ~ 5 s

15. Troubleshooting

Gejala	Tindakan
Mini PC DISCONNECTED	Cek USB Remote↔PC, jalankan exe 2.0, baud 115200
\$WACK TIMEOUT	MAC ESP-NOW, Remote power, jarak
Auto Mini PC tidak gerak	CH6 high, mini_pc_link=1, timestamp CSV cocok dengan balasan
Heading setpoint = yaw	Normal pada alg 2. Bearing hanya ada di alg 1
u, v aneh	Remote belum di-flash yaw CW, atau log lama (skala IMU)
Shutdown tombol abu-abu	Harus Connect + CONNECTED
\$SACK,OK tapi PC tidak mati	Pastikan exe 2.0 yang menangani \$SHUTDOWN
Telemetry 23 kolom	Flash User/Remote-05; dashboard 1.6 tetap terima 23/24

Diagram alur



Dokumen ini: Ship Auto Way Maps Points Tracking — dashboard 1.6, Remote/User-05, Cpp_ReadWriteSerial-2.0-ENU-NMPC-beta